Bishwa Ranjan Routray

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Summary

Aspiring Software Engineer skilled in backend development, REST APIs, and Generative AI applications using Java, Spring Boot, Node.js, PostgreSQL, Python, LangChain, and Gemini API. Strong problem-solving abilities demonstrated through 500+ DSA problems solved and hands-on experience building RAG-based and full-stack applications.

Education

LNCT Group of Colleges — Bhopal, MP
B.Tech in CSE (AIML), CGPA: 8.42 — *2023 – 2027*

Loyola School — Bilaspur, C.G
CBSE Class XII, Percentage: 87 — *2023*

Technical Skills

Languages: C++,Java
Backend: Spring Boot, Spring MVC, Spring Data JPA, Node.js, Express.js, REST APIs
Frontend: HTML, CSS, JavaScript
Databases: MySQL, PostgreSQL
Generative AI: LangChain, Gemini API, ChatGPT API
Concepts: OOP, SOLID Principles, Design Patterns, Microservices Fundamentals

Projects

Ask This Page | *Python, Flask, LangChain, Gemini API, FAISS* — Github link

- Built a Chrome Extension used by **200+** users enabling real-time chat with any webpage or document or YouTube video using **RAG + Gemini API**
- Implemented **RAG (Retrieval-Augmented Generation)** pipeline for context-aware answers and summaries.
- Developed a **Flask backend** handling vector embeddings and Gemini API queries.
- Optimized response accuracy and latency by combining **embedding based retrieval + LLM reasoning** instead of naive prompting.

College Connect | *Node.js, JavaScript, Express, PostgreSQL* — Live Demo

- Developed a **full-stack backend system** that enables college students to register, authenticate, and interact on a coding-focused collaboration platform.
- Implemented **rating and recommendation systems** to highlight reliable and active peers.
- Created **RESTful APIs** and handled database operations with PostgreSQL.
- Integrated **AI chatbot** to resolve queries of the **300+ students** related to academics.

Experience

Machine Learning Intern — Live Demo
Tech:Python,TensorFlow,CNNs,Streamlit — *Edunet Foundation*

- Built a **Convolutional Neural Network (CNN)** to classify plant leaf diseases with approximately **95.2 percent** accuracy.
- Trained and evaluated the model on a labeled dataset of **24 thousand**, applying data preprocessing and model optimization for improved performance.
- Deployed the trained model using **Streamlit** to provide a real-time web interface for disease prediction with under **2 second** interface time.

Certifications

- Generative AI Fundamentals – IBM certificate link
- Machine Learning for All – Coursera certificate link
- Career Essentials in Generative AI – Microsoft and LinkedIn certificate link

Achievements

- LeetCode: Knight Rating **1939**, solved **500+** problems
- Codeforces: **Specialist**, max rating **1567**
- CodeChef: **3-Star**, max rating **1600**